Liv d'Aliberti

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https://liv-daliberti.github.io/



Robust Agentic AI Reasoning and Planning

Education

2024 – Present **Ph.D., Princeton University** Computer Science.
Advisor: Manoel Ribeiro
Affiliations: Center for Information Technology Policy (CITP) | AI Lab | RL Lab

2018 – 2020 **M.S., Georgetown University** in Mathematics & Statistics.

2013 – 2017 **B.S., Georgetown University** in Mathematics.
Honors B.A. in History. Minor: Economics.

Research Publications

Selected Conference Proceedings

1. L. G. d'Aliberti, M. Abdulhai, S. Druchyna, P. Henderson, and M. Horta Ribeiro, "Measuring and mitigating solution mode collapse in RLVR," in *International Conference on Learning Representations (ICLR)*, Accepted at the 6th Workshop on Mathematical Reasoning and AI (MATH-AI), NeurIPS 2026; under review at ICLR 2027, 2027. URL: https://mathai-2026.github.io/.

2. L. G. d'Aliberti, L. Mok, K. Sieck, and M. Horta Ribeiro, "Can language models forecast inductively?" In *International Conference on Learning Representations (ICLR)*, Accepted at SocialAgent: Second Workshop on Large Language Models for Social Reasoning and Simulation, NeurIPS 2026; under review at ICLR 2027, 2027. URL: https://social-llm-workshop.github.io/.

3. M. Hussing, L. G. d'Aliberti, C. Voelcker, B. Eysenbach, and E. Eaton, "Behavior-consistent deep reinforcement learning," in *International Conference on Learning Representations (ICLR)*, Accepted at the Automated RL Workshop (AutoRL), Reinforcement Learning Conference (RLC) 2026; under review at ICLR 2027; arXiv preprint arXiv:2605.21214, 2027. arXiv: 2605.21214 [cs.LG]. URL: https://arxiv.org/abs/2605.21214.

4. L. G. d'Aliberti and M. Horta Ribeiro, "The illusion of insight in reasoning models," in *Findings of the Association for Computational Linguistics: ACL 2026*, Association for Computational Linguistics, 2026, pp. 924–966. DOI: 10.48550/arXiv.2601.00514. arXiv: 2601.00514 [cs.AI].

5. C. A. Clark, K. M. Albarado, J. D. Scott, C. A. Bernard, J. P. Wilson, and L. G. d'Aliberti, "Explainability for UAV control via multi-objective reinforcement learning," in *2025 IEEE Aerospace Conference*, 2025, pp. 1–10. DOI: 10.1109/AERO63441.2025.11068676.

6. L. G. d'Aliberti, E. Gronberg, and J. Kovba, "Privacy-enhancing technologies for AI systems: A tutorial," in *Proceedings of the 10th ACM International Workshop on Security and Privacy Analytics (IWSPA 2024)*, Preprint: arXiv:2404.03509, 2024, pp. 15–26. DOI: 10.1145/3643651.3659889. arXiv: 2404.03509 [cs.CR].

7. V. Miranda and L. G. d'Aliberti, "Hyperdimensional computing encoding schemes for improved image classification," in *2022 IEEE International Symposium on Technologies for Homeland Security (HST 2022)*, 2022, pp. 1–6. DOI: 10.1109/HST56032.2022.10024980.

Workshop Proceedings

1. J. Roth, K. Wagner, and L. G. d'Aliberti, *Label-efficient dark-vessel detection in SAR: Does SAR-domain pretraining outperform optical and ImageNet transfer across ViT and CNN?* Workshop poster; summer student research mentorship, 2026.

2 L. G. d'Aliberti and M. Horta Ribeiro, *Grounded-retrieval adversarial imitation loop (GRAIL)*, Non-archival; accepted. OpenReview id: MdmszyLpVu, 2025. URL: https://openreview.net/forum?id=MdmszyLpVu.

Journal Articles

1 L. G. d'Aliberti and M. Clark, "Preserving patient privacy during computation over shared electronic health record data," *Journal of Medical Systems*, vol. 46, no. 9, p. 87, 2022. DOI: 10.1007/s10916-022-01865-5.

2 S. A. Kareff, L. G. d'Aliberti, and N. Duong, "Effects of a resident-led subject exam review on mean scores of internal medicine subject exam: A case-control study," *Medical Science Educator*, vol. 31, no. 1, pp. 49–52, 2021. DOI: 10.1007/s40670-020-01157-y.

3 S. Woo, K. Johnson, R. Xi, L. G. d'Aliberti, N. Duong, and J. Jennings, "Pre-endoscopic intubation for emergent GI bleeding associated with increased adverse outcomes," *The American Journal of Gastroenterology*, vol. 115, S259, Oct. 2020, Supplement. DOI: 10.14309/01.ajg.0000704332.17755.ab.

Blog Posts

1 J. Miles, L. G. d'Aliberti, C. Renzo, and J. Kovba, *Large language model inference over confidential data using AWS nitro enclaves*, AWS Machine Learning Blog, Blog post, 2024. URL: https://aws.amazon.com/blogs/machine-learning/large-language-model-inference-over-confidential-data-using-aws-nitro-enclaves/.

2 L. G. d'Aliberti and T. Boggs, *Leidos researchers compete to develop the best unmanned surface vessel models*, Leidos Insights Blog, Blog post, 2023. URL: https://www.leidos.com/insights/leidos-researchers-compete-develop-best-unmanned-surface-vessel-model.

3 L. G. d'Aliberti, J. Kovba, M. Gulati, S. Hoda, and B. Snively, *Enabling fully homomorphic encryption with AWS sagemaker endpoints for secure, real-time inferencing*, AWS Machine Learning Blog, Blog post, 2023. URL: https://aws.amazon.com/blogs/machine-learning/enable-fully-homomorphic-encryption-with-amazon-sagemaker-endpoints-for-secure-real-time-inferencing/.

Patents & Invention Disclosures

2024 **System and Process for Securing Client Data During Federated Learning**, U.S. Patent Application No. 20240413969.
Inventors: Olivia Galliker d'Aliberti, Evan Michael Gronberg
Applicant: Leidos Inc.
Filed: June 12, 2024
Published: December 12, 2024.

2026 **Holographic Graph Transformer Network (HGTN) System and Method**, U.S. Patent Application No. 20260087305.
Inventors: Joshua P. Wilson, Jackson D. Scott, Christian A. Clark, Bryce E. King, Olivia Galliker d'Aliberti
Applicant: Leidos, Inc.
Filed: September 24, 2025
Published: March 26, 2026.

Employment History

2019 – 2024 **Senior AI/ML Researcher,** Leidos Inc., Arlington, VA.
• Directed applied-research teams in reinforcement learning, privacy-preserving ML, AutoML, and ML engineering.
• Principal Investigator — Decision-Support RL ($1.2M) and Privacy-Enhancing Technologies ($350K).
• Supported new-business capture efforts totaling ≈$5B and delivery across $7B in active contracts.
• Founded and scaled the AIML Internship Program (5 → 12 interns; 3–5 annual hires).

2023 – Present **Adjunct Lecturer,** Johns Hopkins University, Whiting School of Engineering (Remote; Baltimore, MD).
• Instructor & curriculum designer for EN.605.256 "Modern Software Concepts in Python" (2024–Present) and EN.605.206 "Introduction to Programming Using Python" (2023–2024).
• Instructor for EN.705.643 "Deep Learning Developments with PyTorch" (2025–Present; online synchronous).
• Created hands-on labs integrating real-world AI/ML case studies; taught 50+ graduate students per cohort.
• Consistently earned teaching-evaluation scores $> 4.7/5$.

2017 – 2019 **Data Analyst,** Model B., Washington, D.C.
• Architected cross-platform campaign-analytics pipelines and automated large-scale data ingestion for marketing intelligence.

Skills

Languages English (native); French (professional proficiency).

Programming Python, R, C++.

ML Frameworks PyTorch, TensorFlow, Scikit-Learn, Ray, OpenCV, NumPy, Pandas.

Cloud/DevOps AWS (ML Specialty & Solutions Architect), Docker, Kubernetes, Linux CLI, Git, Neo4j, Weights & Biases.

Web Dev Html, Css, JavaScript, Flask, FastAPI.

Tools LaTeX, JupyterLab, VS Code, Matplotlib.

Miscellaneous Experience

Awards and Achievements

2024 **Purple Dollar Award**. Leidos CTO Office recognition for the first customer transition of an internal R&D program.

2023 **Applied R&D Award**. Leidos CTO Office honor for best department-wide applied research.

Miscellaneous Experience (continued)

- 2021 — **Rising Star Award**. Leidos Annual Achievement Award for leadership and integrity.
- 2021 – 2022 — **AWS DeepRacer Top 10% & Global Finalist**. Ranked in the top decile worldwide and invited to the global finals.
- 2020 — **Object Detection at the Edge Winner**. First place, Leidos AIPalooza competition.
- 2021 — **Judges' Choice & People's Choice**. Varec Hackathon, Leidos subsidiary.

Certification

- 2024 — **AWS Certified Machine Learning – Specialty**. Amazon Web Services.
- 2023 — **AWS Certified Solutions Architect – Associate**. Amazon Web Services.
- 2022 — **Neo4j Certified Professional**. Neo4j, Inc.
- **CompTIA Security+**. Computing Technology Industry Association.
- 2021 — **TensorFlow for AI Professional Certificate**. DeepLearning.AI.

References

Available on Request